

MACHINE LEARNING
PROJECT DOCUMENT

A Structured 4-Sprint Methodology

Objective

To design, build, evaluate, and deploy a machine learning solution using a structured 4-sprint methodology that reflects real-world industry workflows.

Project Structure Overview

Sprint	Focus Area
Sprint 1	Data Understanding & Preprocessing
Sprint 2	Model Building & Evaluation
Sprint 3	Optimization & Final Model
Sprint 4	Deployment & MLOps

SPRINT 1: DATA UNDERSTANDING & PREPROCESSING

Goal: Convert raw data → clean, meaningful, model-ready dataset

Step 1: Data Collection & Loading

- Import dataset from: CSV / Excel / Database / API
- Load using appropriate tools (e.g., pandas)
- Validate:
 - Dataset size
 - Column structure

Step 2: Initial Data Inspection

- Perform:
 - `.head()`, `.tail()`
 - `.info()`
 - `.describe()`
- Check:
 - Data types
 - Missing values
 - Unique values
 - Data consistency

Step 3: Data Cleaning

- Handle missing values: Drop / Impute (mean, median, mode)
- Remove duplicates
- Fix incorrect data types
- Standardize column names

Step 4: Exploratory Data Analysis (EDA)

Univariate Analysis

- Distribution plots
- Identify skewness

Bivariate Analysis

- Feature vs Target relationship
- Correlation analysis

Multivariate Analysis

- Pairplots / heatmaps
- Interaction effects
- Goal: Understand patterns and relationships

Step 5: Outlier Detection & Treatment

- Use: Boxplots, IQR / Z-score
- Actions: Remove / Cap / Transform

Step 6: Feature Encoding

- Convert categorical → numerical:
 - Label Encoding
 - One-Hot Encoding

Step 7: Feature Scaling

- Apply: Standardization / Normalization
- Important: Fit only on training data

Step 8: Train-Test Split

- Split dataset:
 - 70–80% Training
 - 20–30% Testing

Sprint Deliverables

- ✓ Clean dataset
- ✓ EDA report (plots + insights)
- ✓ Preprocessing steps documented

SPRINT 2: MODEL BUILDING & EVALUATION

Goal: Build baseline and compare multiple models

Step 1: Baseline Model

- Train a simple model:
 - Regression → Linear Regression
 - Classification → Logistic Regression
- Purpose: Establish performance baseline

Step 2: Train Multiple Models

For Regression:

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting

For Classification:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- Naive Bayes

Step 3: Model Evaluation

Regression Metrics:

- MAE
- RMSE
- R^2

Classification Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Step 4: Overfitting & Underfitting Check

- Compare: Training vs Testing performance

Step 5: Model Comparison Table

Model	Metric Score	Overfitting?	Remarks
Linear Regression	—	—	—
Decision Tree	—	—	—
Random Forest	—	—	—
Gradient Boosting	—	—	—

Sprint Deliverables

- ✓ Multiple trained models
- ✓ Evaluation metrics
- ✓ Best performing baseline model

SPRINT 3: OPTIMIZATION & FINAL MODEL

Goal: Improve model performance and reliability

Step 1: Feature Engineering

- Create new features:
 - Ratios
 - Aggregations
 - Domain-specific transformations

Step 2: Feature Selection

- Remove: Irrelevant features, Highly correlated features
- Methods:
 - Correlation matrix
 - Feature importance
 - Recursive Feature Elimination (RFE)

Step 3: Hyperparameter Tuning

- Use: GridSearchCV, RandomizedSearchCV
- Tune: Model-specific parameters

Step 4: Build Final Model

- Train model using:
 - Best features
 - Best hyperparameters

Step 5: Final Evaluation

- Evaluate on unseen test data

Step 6: Model Serialization

- Save model: .pkl or .joblib

Sprint Deliverables

- ✓ Final optimized model
- ✓ Feature engineering steps
- ✓ Saved model file
- ✓ Interpretation insights

SPRINT 4: DEPLOYMENT & MLOPS

Goal: Make model usable in real-world applications

Step 1: Build ML Pipeline

- Combine: Preprocessing + Model
- Use pipeline structure

Step 2: Frontend

- Build UI using Streamlit
- Features:
 - User inputs
 - Prediction display

Step 3: Deployment

- Deploy model: Local / Cloud platforms

Step 4: MLOps Practices

Version Control

- Use GitHub

Experiment Tracking

- Track: Models, Metrics, Parameters

Logging & Monitoring

- Log predictions
- Monitor model performance

Step 5: Project Structure

```
project/
├── data/
├── notebooks/
├── src/
│   ├── preprocessing.py
│   ├── train.py
│   └── predict.py
├── models/
│   └── model.pkl
├── app/
│   └── app.py
├── requirements.txt
└── README.md
```

Step 6: Documentation

- Include:
 - Problem statement
 - Approach
 - Results
 - How to run project

Sprint Deliverables

- ✓ Deployed model (API/UI)
- ✓ End-to-end pipeline
- ✓ GitHub repository
- ✓ Documentation